Diagonal concavity does not imply concavity for sums of elementary symmetric polynomials

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26 October 2026

Abstract

An explicit family disproves Lachaume's diagonal concavity conjecture in four variables with strictly positive coefficients. For the family $f_M = 1 + M\sigma_1 + \sigma_2 + M\sigma_3 + \sigma_4$, where $M > 0$, the fourth root is concave on the positive diagonal exactly when $M \geq 1$. It is concave on the full positive orthant exactly when $M = 1$. Two polynomial identities prove the failure of concavity. The remaining positive case follows from the arithmetic–geometric mean inequality. The accompanying Lean development proves both classifications with the usual concavity predicate and canonical elementary symmetric functions.

1 Statement

For $x = (x_1, x_2, x_3, x_4)$, define the elementary symmetric polynomials by

$$\sigma_j(x) = \sum_{1 \leq i_1 < \cdots < i_j \leq 4} x_{i_1} \cdots x_{i_j}, \qquad \sigma_0(x) = 1.$$

Write $\Gamma = (0, \infty)^4$. A function H is concave on a convex set if

$$H(\theta x + (1-\theta)y) \geq \theta H(x) + (1-\theta)H(y) \quad (0 \leq \theta \leq 1)$$

for all x, y in that set. Lachaume's Conjecture 1 [2] asserts an equivalence between concavity on the positive orthant and concavity on its positive diagonal for

$$\left(\sum_{j=0}^{p} a_j \sigma_j \right)^{1/p}, \qquad a_j \geq 0.$$

Here the positive diagonal consists of the vectors $(t, \ldots, t)$ with $t > 0$.

Theorem 1. *For $M > 0$, put*

$$f_M = 1 + M\sigma_1 + \sigma_2 + M\sigma_3 + \sigma_4, \qquad H_M = f_M^{1/4}.$$

Then

$$\begin{aligned} t \longmapsto H_M(t,t,t,t) \text{ is concave on } (0,\infty) &\iff M \geq 1, \\ H_M \text{ is concave on } \Gamma &\iff M = 1. \end{aligned}$$

Thus each $M > 1$ disproves Conjecture 1. In particular, $(a_0, a_1, a_2, a_3, a_4) = (1, 2, 1, 2, 1)$ is a counterexample with all coefficients positive.

All terms in f_M are positive or nonnegative on Γ, and its constant term is one. Thus its fourth root is defined and smooth throughout the domain. The polynomial has degree four. No zero coefficient or boundary point is needed.

2 Two curvature identities

For a positive twice differentiable function q,

$$(q^{1/4})'' = \frac{4qq'' - 3(q')^2}{16q^{7/4}}. \tag{1}$$

A twice differentiable function on an open interval is concave exactly when its second derivative is nonpositive. This follows from the mean value theorem and the monotonicity of its first derivative.

The diagonal polynomial is

$$P_M(t) = f_M(t,t,t,t) = 1 + 4Mt + 6t^2 + 4Mt^3 + t^4.$$

Expansion gives

$$3(P_M'(t))^2 - 4P_M(t)P_M''(t) = 48(M^2-1)(t^2-1)^2. \tag{2}$$

If $M \geq 1$, equations (1) and (2) give concavity on the diagonal. If $0 < M < 1$, the second derivative is positive for every $t > 0$ except $t = 1$. This proves the first classification. It also excludes full concavity when $0 < M < 1$, because restriction to an affine line preserves concavity.

To treat $M > 1$, let

$$x_* = (2, 2, 1/2, 1/2), \qquad v = (2, 2, 1, 1), \qquad Q_M(s) = f_M(x_* + sv).$$

This line lies in Γ for $s > -1/2$. Direct expansion gives

$$\begin{aligned} Q_M(s) = 4s^4 + (12M+12)s^3 + (28M+26)s^2 \\ + 27(M+1)s + (41+40M)/4. \end{aligned}$$

Thus

$$Q_M(0) = \frac{41+40M}{4}, \qquad Q_M'(0) = 27(M+1), \qquad Q_M''(0) = 52 + 56M.$$

It follows that

$$4Q_M(0)Q_M''(0) - 3(Q_M'(0))^2 = (M-1)(53M+55) > 0. \tag{3}$$

Equation (1) gives a positive second derivative of H_M along this interior line. Hence H_M is not concave when $M > 1$.

3 The positive case

It remains to prove concavity when $M = 1$. The subset expansion gives

$$f_1(x) = \prod_{i=1}^{4}(1 + x_i).$$

The required geometric-mean concavity has the following elementary proof. For $u \in (0,\infty)^4$, put $G(u) = (u_1u_2u_3u_4)^{1/4}$. The arithmetic–geometric mean inequality implies, for positive u, z,

$$G(u) \leq \frac{G(z)}{4}\sum_{i=1}^{4}\frac{u_i}{z_i}. \tag{4}$$

For positive u, w, take $z = \theta u + (1-\theta)w$, where $0 \leq \theta \leq 1$. Apply (4) to u and w, multiply by their weights, and add. This yields

$$\theta G(u) + (1-\theta)G(w) \leq \frac{G(z)}{4}\sum_{i=1}^{4}\frac{\theta u_i + (1-\theta)w_i}{z_i} = G(z).$$

Thus G is concave. Composition with the affine map $x \mapsto (1+x_1, \ldots, 1+x_4)$ proves concavity of H_1. Together with the preceding section, this proves Theorem 1.

4 A finite midpoint witness

An independent certificate shows the failure at $M = 2$ directly from the defining concavity inequality. Let

$$x_- = (3/2, 3/2, 1/4, 1/4), \qquad x_+ = (5/2, 5/2, 3/4, 3/4).$$

Their midpoint is x_*. Exact evaluation gives

$$f_2(x_-) = 933/64, \quad f_2(x_*) = 121/4, \quad f_2(x_+) = 3597/64.$$

The rational comparisons

$$(977/500)^4 < 933/64, \qquad (1369/500)^4 < 3597/64, \qquad 121/4 < (4691/2000)^4$$

give

$$\frac{H_2(x_-) + H_2(x_+)}{2} > \frac{1173}{500} > \frac{4691}{2000} > H_2(x_*).$$

This is the strict reverse of the required midpoint inequality.

5 Formal correspondence and scope

The Lean namespace `MathAndTheory.DiagonalConcavity` proves the polynomial identities, derivatives, and analytic implications. The declaration `complete_classification` gives Theorem 1. The definition `elementary` uses Mathlib's `Multiset.esymm`; the theorem `family_eq_elementary` connects it to the coordinate formulas. The declaration `elementary_symmetric_counterexample` states the result using those canonical symmetric functions. The declaration `midpoint_strict_violation` gives the separate midpoint certificate.

The result addresses Conjecture 1. It does not classify arbitrary coefficient vectors.

Lachaume's separate sufficient condition requires a real-rooted diagonal polynomial. The present result does not contradict that condition.

The Walsh-convolution note [1] concerns Conjectures 4 and 5. It makes no claim for Conjectures 1 and 3. The inspected sources supplied no earlier disproof of Conjecture 1. This bounded search does not establish first priority.

The original research and Lean development were generated by ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning, under the direction of Dev D. Goyal. This revision includes further AI-assisted checking and editing. Machine verification and AI review do not constitute independent expert peer review. This work has not been peer reviewed.

References

[1] Carptopus. A walsh-convolution proof of two conjectures of lachaume, September 2026. URL https://github.com/Carptopus/MathExplore-Notes/blob/6e96176abbbb884906b2235b418d2315908bb5bd/research/lachaume-walsh-convolution/README.md. Public version dated 4 September 2026. The source states that it addresses Conjectures 4 and 5, not Conjectures 1 and 3.

[2] Xavier Lachaume. On the concavity of a sum of elementary symmetric polynomials, 2017. URL https://arxiv.org/abs/1712.10327v1. Conjecture 1; submitted 12 December 2017.